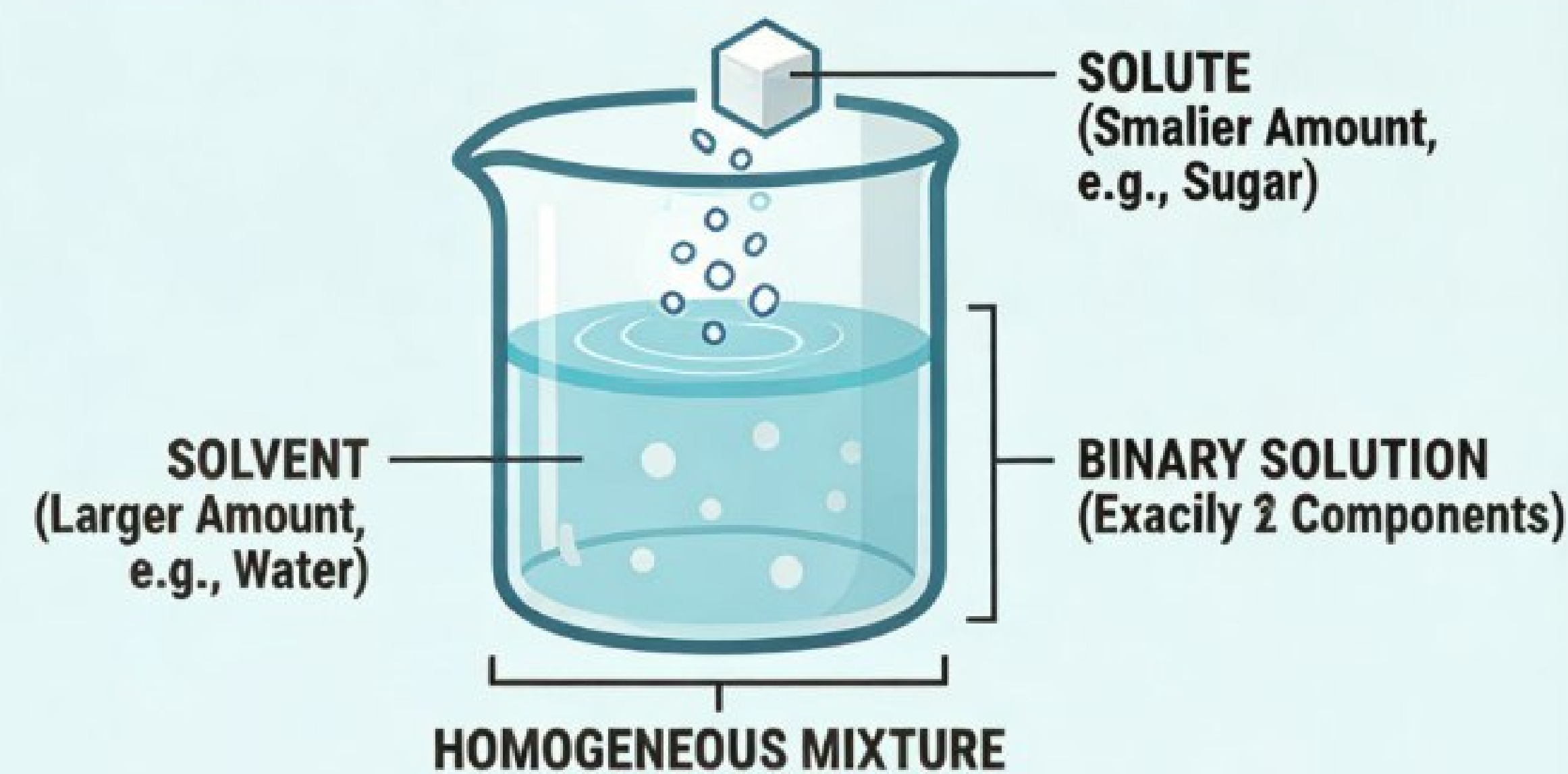


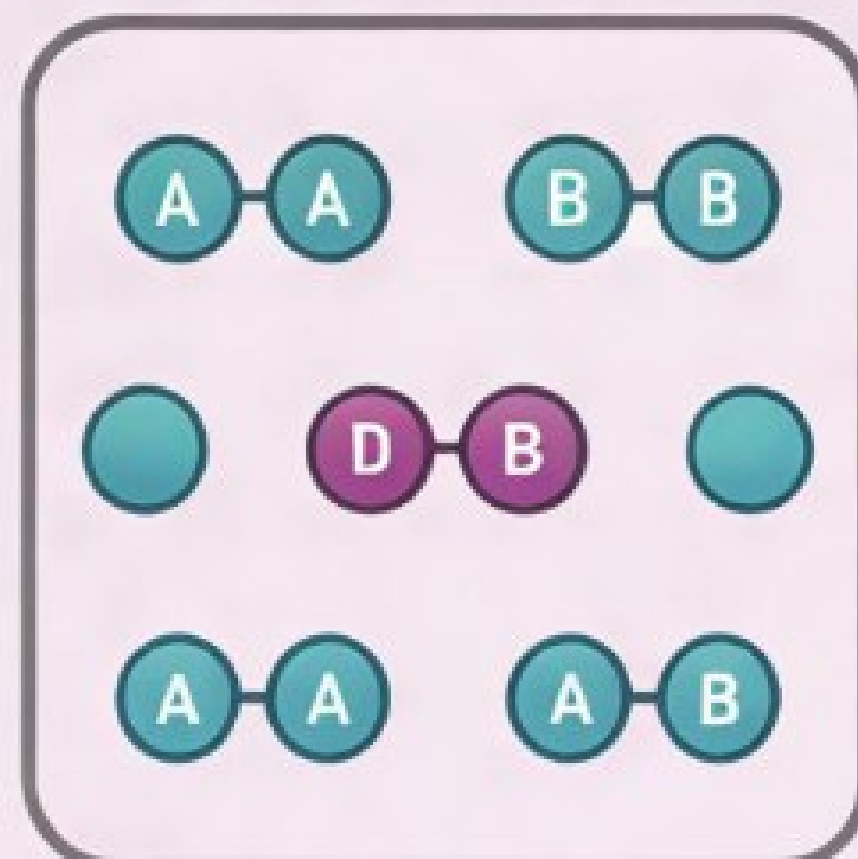
A Complete Guide to Chemical Solutions

THE BASICS OF A SOLUTION

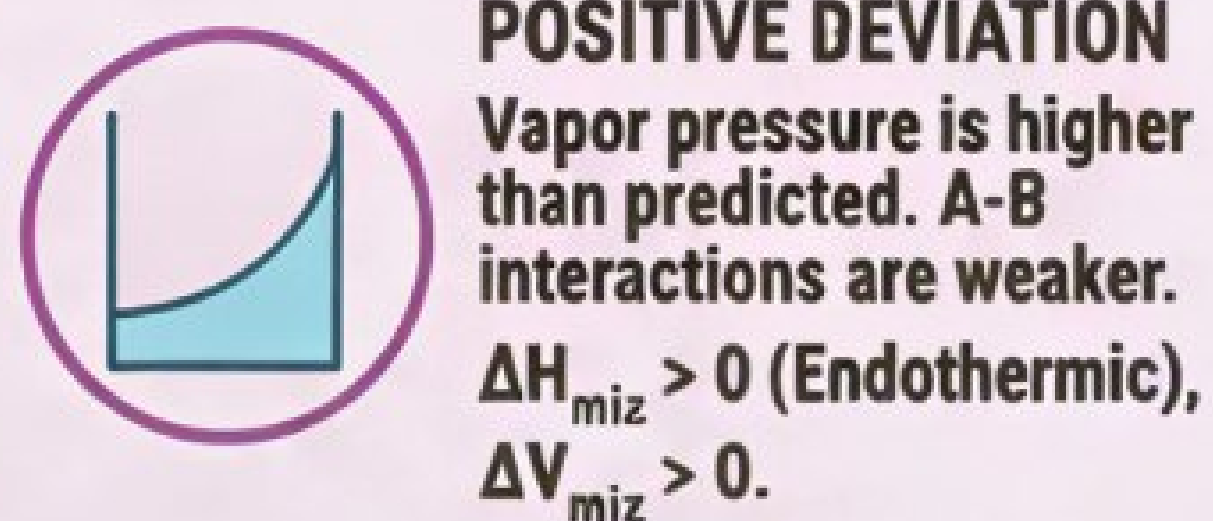


IDEAL vs. NON-IDEAL SOLUTIONS

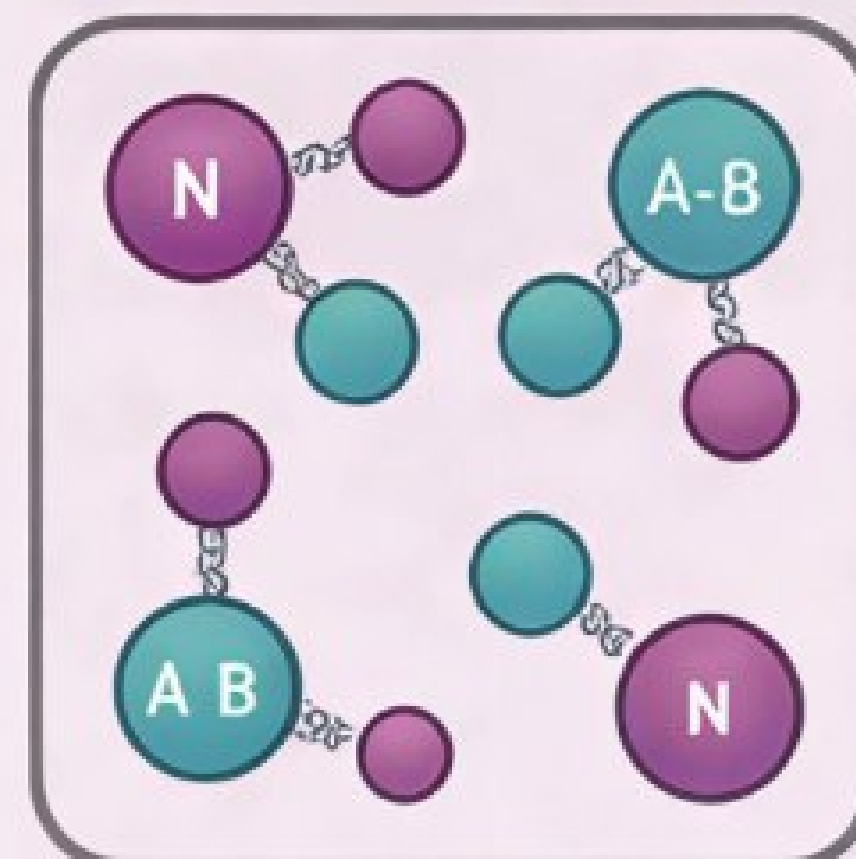
IDEAL SOLUTIONS



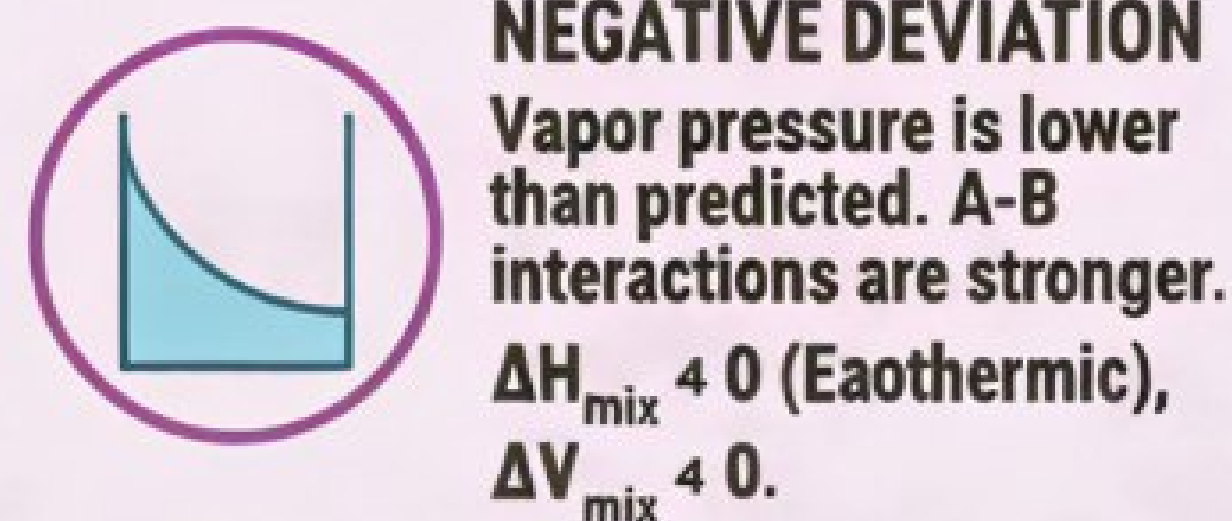
✓ Obey Raoult's Law.
 $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$.



NON-IDEAL SOLUTIONS



✗ Do not obey Raoult's Law.
Interactions are unequal.
 $\Delta H_{mix} \neq 0$ and $\Delta V_{mix} \neq 0$.



MEASURING CONCENTRATION



MASS PERCENTAGE (%)
 $\left(\frac{\text{Mass of Solute}}{\text{Mass of Solution}} \right) \times 100$



PARTS PER MILLION (PPM)
 $\left(\frac{\text{Mass of Component}}{\text{Mass of Solution}} \right) \times 10^6$



MOLARITY (M)
 $\frac{\text{Moles of Solute}}{\text{Volume of Solution (in Liters)}}$



MOLALITY (m)
 $\frac{\text{Moles of Solute}}{\text{Mass of Solvent (in kg)}}$

Note: Molarity is dependent on temperature.

Note: Molality is independent of temperature.

FUNDAMENTAL LAWS & CONCEPTS



MOLE FRACTION (X)
 $\frac{\text{Moles of a Component}}{\text{Total Moles in Solution}}$
For a binary solution,
 $X_A + X_B = 1$.



EFFECT OF TEMPERATURE ON SOLUBILITY:
• Endothermic processes: Solubility increases with temperature.
• Exothermic processes: Solubility decreases with temperature.



HENRY'S LAW
The partial pressure of a gas in the vapor phase (P) is directly proportional to its mole fraction (X) in the solution.
 $P = K_H \cdot X$



SOLUBILITY
The maximum amount of solute that can be dissolved in a specific amount of solvent at a particular temperature.

CORRECTING FOR REALITY: ABNORMAL MOLAR MASS

When the molar mass calculated from colligative properties is either lower or higher than the expected normal value. This occurs when solutes associate or dissociate.



COLLIGATIVE PROPERTIES
Properties of a solution that depend only on the number of solute particles, not on their nature or identity.



RELATIVE LOWERING IN VAPOUR PRESSURE

$\frac{P^\circ - P_1}{P^\circ} = X_2$
Where P° is the vapor pressure of the pure solvent.



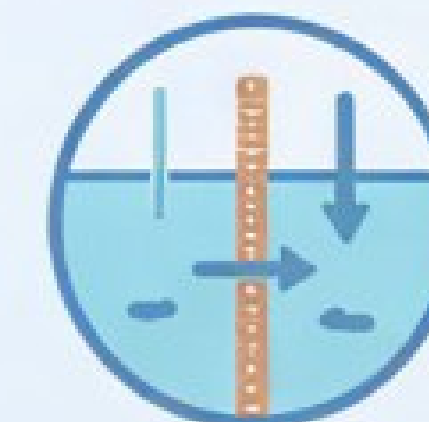
DEPRESSION IN FREEZING POINT

$\Delta T_f = K_f \cdot m$
The freezing point of a solution with a non-volatile solute is lower than the pure solvent.



ELEVATION IN BOILING POINT

$\Delta T_b = K_b \cdot m$
The boiling point of a solution with a non-volatile solute is higher than the pure solvent.

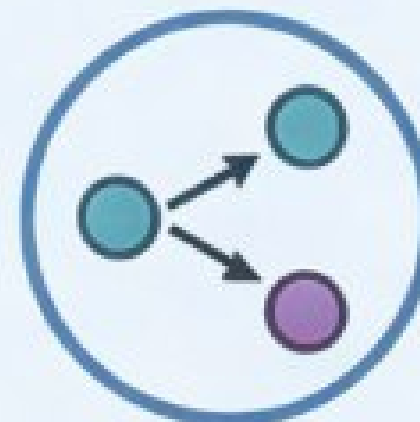


OSMOTIC PRESSURE (π)

$\pi = CRT$
This is the pressure required to stop the process of osmosis (flow of solvent from low to high concentration across a semipermeable membrane).

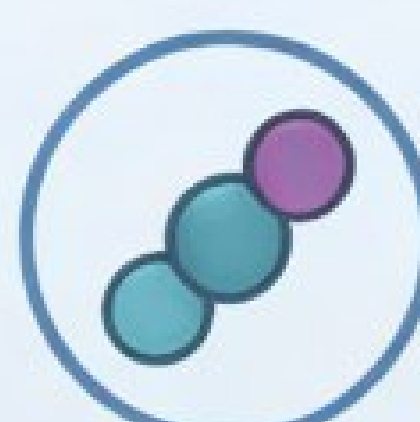


VAN'T HOFF FACTOR (i)
A correction factor applied to colligative property formulas to account for the association or dissociation of solute particles.



CALCULATING 'i' FOR DISSOCIATION

$i = 1 + (n - 1)\alpha$,
where 'n' is the number of particles formed and 'α' is the degree of dissociation.



CALCULATING 'i' FOR ASSOCIATION

$i = 1 + \left(\frac{1}{n} - 1 \right) \alpha$
where 'n' is the number of particles associating and 'α' is the degree of association.